

Практикум

D4:archive

Twenty-four questions, 104 marks. Every question the archive asks about a discrete random variable that is not binomial, from May 2021 to November 2025, in the order of the six techniques rather than the order of the papers.

О практикуме

Собран из трёх тем: 50 баллов

statistics.discrete_random_variables (0.562 — 50 из 89), 48 statistics.expected_value (0.800 — 48 из 60) и 11 производящей функции из statistics.probability (0.046 — 11 из 237). 109 баллов по корпусу, 104 рабочих: пять — ноябрьский дубль 2023 года. Первая тема статистики, стоящая на трёх бумагах почти поровну: 41 балл Paper 1, 30 Paper 2, 33 Paper 3.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-d4-discrete

Собрано из ноутбука репозитория [ib-math-aa-hl-past-papers](#). Проверки в PDF не работают — они для интерактивной версии.

D4 archive — discrete random variables, all of it

Twenty-four questions, 104 marks.

Every question the archive asks about a discrete random variable that is not binomial, from May 2021 to November 2025, in the order of the six techniques rather than the order of the papers.

No theory. No worked examples. The theory is in the practicum, `practicum-d4-discrete.ipynb`.

§	technique	questions	marks
1	The table must be a distribution	4	14
2	The mean	6	19
3	Two letters, two conditions	5	28
4	Spread, and a variable built from X	4	16
5	A first success	2	9
6	A generating function	3	18

The checks are the same ones the practicum uses and they store nothing: each is handed the table and the conditions of the question, finds the letters itself, and throws away the roots that break a cell. Exact values on Paper 1 questions — `Rational(2, 7)`, `Interval(0, Rational(1, 3))` — and three significant figures elsewhere.

Parts of one question are split by technique. May 2021 TZ1 Q10 appears in §§ 1, 2 and 3; each part repeats the table it needs.

The November 2023 paper appears once, although the archive holds it twice as TZ1 and TZ2.

Solutions are at the very bottom, deliberately far away.

```
import sys
sys.path.append('..')          # from
practicum/statistics to practicum/kit/
import sympy as sp             # the escape
hatch: anything not in kit is in sp
from kit import *              # checks +
Dist, Freq, Geo, Moments, P(), Expect, Var,
Pgf

language('en')                 # this
notebook is in English, and so are the checks

print('ready; sympy', sp.__version__)
```

§ 1. The table must be a distribution

Four questions, 14 marks. Every cell in $[0, 1]$, the cells add up to one — and a root that breaks a cell is rejected, with the reason.

1.1 — May 2025 TZ3 Paper 2 Q3(a), 3 marks

The number of times, X , each customer visits a supermarket in a week is given by the following probability distribution.

x	1	2	3	4	5	≥ 6
$P(X = x)$	$1.5a$	$2a$	0.281	a	0.026	0

(i) Find the value of a . **(ii)** Write down the mode of X .

```
q1_la = ...      # a
q1_lm = ...      # the mode

a = symbols('a')
visits = Dist({1: 1.5*a, 2: 2*a, 3: 0.281, 4: a, 5: 0.026, 6: 0})

verify_letters('1.1(i)', q1_la, a, [visits])
verify_mode('1.1(ii)', q1_lm, visits, given=
[], var=a)
```

1.2 — May 2022 TZ1 Paper 2 Q3(a)-(b), 4 marks

A discrete random variable, X , has the following probability distribution:

x	0	1	2	3
$P(X = x)$	0.41	$k - 0.28$	0.46	$0.29 - 2k^2$

(a) Show that $2k^2 - k + 0.12 = 0$. **(b)** Find the value of k , giving a reason for your answer.

```
k = symbols('k')

q1_2e = ...      # the equation, as Eq(...)
q1_2k = ...      # k

X = Dist({0: 0.41, 1: k - 0.28, 2: 0.46, 3:
0.29 - 2*k**2})

verify_equation('1.2(a)', q1_2e,
Eq(total_probability(X), 1), var=k)
verify_letters('1.2(b)', q1_2k, k, [X])
```

1.3 — May 2021 TZ1 Paper 1 Q10(a), (c), 5 marks

A biased four-sided die, A , is rolled, and X is the score.

x	1	2	3	4
$P(X = x)$	p	p	p	$\frac{1}{2}p$

(a) Find the value of p .

A second biased four-sided die, B , is rolled, and Y is the score.

y	1	2	3	4
$P(Y = y)$	q	q	q	r

(c) (i) State the range of possible values of r . **(ii)** Hence, find the range of possible values of q .

```
q1_3p = ...      # p, exactly
q1_3r = ...      # the range of r, as an
Interval
q1_3q = ...      # the range of q

p, q, r = symbols('p q r')
X = Dist({1: p, 2: p, 3: p, 4: p/2}, 'X')
Y = Dist({1: q, 2: q, 3: q, 4: r}, 'Y')

verify_letters('1.3(a)', q1_3p, p, [X])
verify_table_range('1.3(c)(i)', q1_3r, r,
[Y])
verify_table_range('1.3(c)(ii)', q1_3q, q,
[Y])
```

1.4 — November 2025 TZ3 Paper 1 Q7(a), 2 marks

The discrete random variable X has the following probability distribution:

x	1	2	3
$P(X = x)$	$0.6 - 2a$	$3a$	$0.4 - a$

Find the range of possible values of a .

```
q1_4 = ...          # the range of a

a = symbols('a')
X = Dist({1: 0.6 - 2*a, 2: 3*a, 3: 0.4 - a})

verify_table_range('1.4', q1_4, a, [X])
```

§ 2. The mean

Six questions, 19 marks. Each value times its own probability.

2.1 — May 2025 TZ3 Paper 2 Q3(b), 2 marks

(The supermarket of 1.1.) Find the mean of X .

2.2 — May 2022 TZ1 Paper 2 Q3(c), 2 marks

(The table of 1.2.) Hence, find $E(X)$.

```
q2_1 = ...          # the mean number of visits
q2_2 = ...          # E(X) for the table of 1.2

a, k = symbols('a k')
visits = Dist({1: 1.5*a, 2: 2*a, 3: 0.281, 4:
a, 5: 0.026, 6: 0})
X = Dist({0: 0.41, 1: k - 0.28, 2: 0.46, 3:
0.29 - 2*k**2})

verify_moment('2.1', q2_1, Expect(visits),
given=[], var=a)
verify_moment('2.2', q2_2, Expect(X), given=
[], var=k)
```

2.3 — May 2021 TZ1 Paper 1 Q10(b), (d), 5 marks

(The dice A and B of 1.3.)

(b) Hence, find the value of $E(X)$.

(d) Hence, find the range of possible values for $E(Y)$.

```
q2_3b = ...      # E(X), exactly
q2_3d = ...      # the range of E(Y), as an
Interval

p, q, r = symbols('p q r')
X = Dist({1: p, 2: p, 3: p, 4: p/2}, 'X')
Y = Dist({1: q, 2: q, 3: q, 4: r}, 'Y')

verify_moment('2.3(b)', q2_3b, Expect(X),
given=[], var=p)
verify_table_range('2.3(d)', q2_3d,
Expect(Y), [Y])
```

2.4 — May 2023 TZ1 Paper 1 Q2, 6 marks

On a Monday at an amusement park, a sample of 40 visitors was randomly selected as they were leaving the park. They were asked how many times that day they had been on a ride called The Dragon.

number of times on The Dragon	0	1	2	3	4
frequency	6	16	13	2	3

It can be assumed that this sample is representative of all visitors to the park for the following day.

(a) For the following day, Tuesday, estimate **(i)** the probability that a randomly selected visitor will ride The Dragon; **(ii)** the expected number of times a visitor will ride The Dragon.

It is known that 1000 visitors will attend the amusement park on Tuesday. The Dragon can carry a maximum of 10 people each time it runs.

(b) Estimate the minimum number of times The Dragon must run to satisfy demand.

```

q2_4p = ...      # P(a visitor rides)
q2_4e = ...      # the expected number of
rides per visitor
q2_4r = ...      # the minimum number of runs

rides = Freq({0: 6, 1: 16, 2: 13, 3: 2, 4:
3})

verify_chance('2.4(a)(i)', q2_4p, P(rides >=
1))
verify_moment('2.4(a)(ii)', q2_4e,
Expect(rides))
verify_moment('2.4(b)', q2_4r, Expect(rides *
1000 / 10))

```

2.5 — May 2025 TZ1 Paper 3 Q1(a), 2 marks

Two unbiased tetrahedral dice with faces labelled 1, 2, 3 and 4 are thrown, and M is the maximum of the two scores.

m	1	2	3	4
$P(M = m)$	$\frac{1}{16}$	$\frac{3}{16}$	$\frac{5}{16}$	$\frac{7}{16}$

Find $E(M)$.

2.6 — May 2024 TZ1 Paper 3 Q1(g)(ii), 2 marks

In a computer game, Y is the number of actions until the first boost occurs, with

y	1	2	3	4	5
$P(Y = y)$	0.2	0.32	0.288	0.1536	0.0384

Show that $E(Y) = 2.5104$.

```
q2_5 = ...          # E(M)
q2_6 = ...          # E(Y)

throws = {(i, j): Rational(1, 16) for i in
range(1, 5) for j in range(1, 5)}
M = Dist(throws, 'M', rule=max)
Y = Dist({1: 0.2, 2: 0.32, 3: 0.288, 4:
0.1536, 5: 0.0384}, 'Y')

verify_moment('2.5', q2_5, Expect(M))
verify_moment('2.6', q2_6, Expect(Y))
```

§ 3. Two letters, two conditions

Five questions, 28 marks. One equation from the sum, one from the fact the question adds — then the roots the table allows.

3.1 — *May 2022 TZ2 Paper 1 Q10(a), 5 marks*

A biased four-sided die is rolled, and X is the result.

x	1	2	3	4
$P(X = x)$	p	0.3	q	0.1

It is known that $E(X) = 2$. Show that $p = 0.4$ and $q = 0.2$.

3.2 — *November 2023 TZ1 Paper 2 Q5, 5 marks*

The following table shows the probability distribution of a discrete random variable X , where $a, k \in \mathbb{R}^+$.

x	1	2	3	4
$P(X = x)$	k	k^2	a	k^3

Given that $E(X) = 2.3$, find the value of a .

```
q3_1 = [...]      # [p, q]
q3_2 = ...         # a

p, q = symbols('p q')
X = Dist({1: p, 2: 0.3, 3: q, 4: 0.1})
a, k = symbols('a k')
W = Dist({1: k, 2: k**2, 3: a, 4: k**3})

verify_letters('3.1', q3_1, [p, q], [X],
[Eq(Expect(X), 2)])
verify_letters('3.2', q3_2, a, [W],
[Eq(Expect(W), 2.3)])
```

3.3 — May 2024 TZ1 Paper 1 Q1(a), 5 marks

Claire rolls a six-sided die 16 times. The scores obtained are shown in the following frequency table.

score	1	2	3	4	5	6
frequency	p	q	4	2	0	3

It is given that the mean score is 3. Find the value of p and the value of q .

3.4 — May 2025 TZ2 Paper 2 Q8, 7 marks

The marks obtained by students in a class quiz are shown in the following

table where $p, q \in \mathbb{Z}^+$.

marks	20	35	p
frequency	12	q	8

The mean and variance of the marks are 31 and 124 respectively. Find the value of p and the value of q .

```
q3_3 = [...]      # [p, q]
q3_4 = [...]      # [p, q]

p, q = symbols('p q', integer=True,
nonnegative=True)
rolls = Freq({1: p, 2: q, 3: 4, 4: 2, 5: 0,
6: 3})
verify_letters('3.3', q3_3, [p, q], [rolls],
[Eq(rolls.size, 16), Eq(Expect(rolls), 3)])

p, q = symbols('p q', positive=True,
integer=True)
quiz = Freq({20: 12, 35: q, p: 8})
verify_letters('3.4', q3_4, [p, q], [quiz],
[Eq(Expect(quiz), 31), Eq(Var(quiz), 124)])
```

3.5 — May 2021 TZ1 Paper 1 Q10(e), 6 marks

(The dice A and B of 1.3.) Agnes and Barbara play a game using these dice. Agnes rolls die A once and Barbara rolls

die B once. The probability that Agnes' score is less than Barbara's score is $\frac{1}{2}$.

Find the value of $E(Y)$.

```
q3_5qr = [...]    # [q, r]
q3_5 = ...         # E(Y)

p, q, r = symbols('p q r')
X = Dist({1: p, 2: p, 3: p, 4: p/2}, 'X')
Y = Dist({1: q, 2: q, 3: q, 4: r}, 'Y')
game = [Eq(P(X < Y), Rational(1, 2))]

verify_letters('3.5 q, r', q3_5qr, [q, r],
[X, Y], game)
verify_moment('3.5', q3_5, Expect(Y),
given=game, var=[p, q, r], tables=[X])
```

§ 4. Spread, and a variable built from X

Four questions, 16 marks. $E(X^2) - E(X)^2$; the constant goes, the multiplier is squared.

4.1 — November 2025 TZ3 Paper 1 Q7(b), 6 marks

(The table of 1.4.) In the case where $a = 0.2$, determine $\text{Var}(2 - X)$.

4.2 — May 2024 TZ1 Paper 1 Q1(b), 1 mark

(Claire's scores of 3.3.) Each of Claire's scores is multiplied by 10 in order to determine the final score for a game she is playing. Write down the mean final score.

```

q4_1 = ...          # Var(2 - X)
q4_2 = ...          # the mean final score

a = symbols('a')
X = Dist({1: 0.6 - 2*a, 2: 3*a, 3: 0.4 - a})
p, q = symbols('p q', integer=True,
nonnegative=True)
rolls = Freq({1: p, 2: q, 3: 4, 4: 2, 5: 0,
6: 3})
claire = [Eq(rolls.size, 16),
Eq(Expect(rolls), 3)]

verify_moment('4.1', q4_1, Var(2 - X),
given=Eq(a, 0.2), var=a)
verify_moment('4.2', q4_2, Expect(10 *
rolls), given=claire, var=[p, q])

```

4.3 — May 2025 TZ1 Paper 2 Q11(e)-(f), 7 marks

The random variable T is the time, in hours, taken for a runner to complete a marathon. No runner completes the race in less than 2.25 hours. Each runner's time is converted to a score $a - bt$, where t is their time in hours and $a, b > 0$, and P is the score of a runner. It is given that $E(P) = 100$ and the maximum possible score is 150.

(e) Use $E(T) = 4.723$ to determine the value of a and the value of b , giving your answers to the nearest integer.

(f) Given also that $\text{Var}(T) = 0.906$, find $\text{Var}(P)$.

**4.4 — May 2024 TZ1 Paper 3 Q1(g)
(iii), 2 marks**

(The table of Y in 2.6.) Find $\text{Var}(Y)$.

```
q4_3ab = [...]    # [a, b]
q4_3v = ...        # the variance of the score
q4_4 = ...         # Var(Y)

a, b = symbols('a b', positive=True)
T = Moments(4.723, 0.906, 'T')
score = [Eq(Expect(a - b*T), 100), Eq(a -
b*2.25, 150)]
Y = Dist({1: 0.2, 2: 0.32, 3: 0.288, 4:
0.1536, 5: 0.0384}, 'Y')

verify_letters('4.3(e)', q4_3ab, [a, b], [T],
score, whole=True)
verify_moment('4.3(f)', q4_3v, Var(a - b*T),
given=score, var=[a, b])
verify_moment('4.4', q4_4, Var(Y))
```

§ 5. A first success

Two questions, 9 marks. Values $1, 2, 3, \dots$ with no end — the mean is still a weighted sum.

5.1 — May 2024 TZ1 Paper 3 Q1(b)-(d), 6 marks

Each time a player performs an action, the probability that it is *boosted* is p , where $0 < p < 1$, independently.

(b) (i) Explain why the probability that the first boost occurs on the x th action is $p(1-p)^{x-1}$. Let X be the number of actions until the first boost occurs. **(ii)** Hence, write down an expression, using sigma notation, for $E(X)$ in terms of x and p .

(c) (ii) Given that $\sum_{n=1}^{\infty} nar^{n-1} = \frac{a}{(1-r)^2}$, show that $E(X) = \frac{1}{p}$.

It can be shown that $\text{Var}(X) = \frac{1-p}{p^2}$.

(d) Find $E(X)$ and $\text{Var}(X)$ when $p = 0.1$.

```

p = symbols('p')

q5_1b = ...      # E(X) as a sum, with
Sum(..., (x, 1, oo))
q5_1m = ...      # E(X) when p = 0.1
q5_1v = ...      # Var(X) when p = 0.1

verify_moment('5.1(b)(ii)', q5_1b,
Expect(Geo(p)))
verify_moment('5.1(d) E(X)', q5_1m,
Expect(Geo(0.1)))
verify_moment('5.1(d) Var(X)', q5_1v,
Var(Geo(0.1)))

```

5.2 — May 2024 TZ1 Paper 3 Q1(h), 3 marks

(The first model of 5.1, and the second model Y of 2.6 with $E(Y) = 2.5104$ and $\text{Var}(Y) = 1.18749\dots$)

(i) Use $E(X) = \frac{1}{p}$ to find the value of p for which $E(X) = E(Y)$. **(ii)** Find $\text{Var}(X)$ for this value of p . **(iii)** Hence determine, with a reason, which model provides a more consistent experience for the player with respect to boosted actions.

```
q5_2p = ...      # p
q5_2v = ...      # Var(X)

p = symbols('p')
X = Geo(p, 'X')
Y = Dist({1: 0.2, 2: 0.32, 3: 0.288, 4:
0.1536, 5: 0.0384}, 'Y')
same_mean = Eq(Expect(X), Expect(Y))

verify_letters('5.2(i)', q5_2p, p, [X],
[same_mean])
verify_moment('5.2(ii)', q5_2v, Var(X),
given=same_mean, var=p)
```

§ 6. A generating function

Three questions, 18 marks. The coefficient of t^x is $P(X = x)$.

6.1 — May 2025 TZ1 Paper 3 Q1(c) (ii), 3 marks

(The maximum M of 2.5.) The distribution of M is represented by $G(t) = \sum_{m=1}^4 P(M = m)t^m = \frac{1}{16}t + \frac{3}{16}t^2 + \frac{5}{16}t^3 + \frac{7}{16}t^4$. Part (c)(i) finds $G'(t)$. Hence, show that $G'(1) = E(M)$.

6.2 — May 2025 TZ1 Paper 3 Q1(d), 5 marks

A bag contains two red balls and three yellow balls. Two balls are selected at random without replacement. X is the total number of red balls selected, and $G_X(t) = \sum_{x=0}^2 P(X = x)t^x$. Show that $G_X(t) = \frac{3}{10} + \frac{3}{5}t + \frac{1}{10}t^2$, making it clear how the coefficients have been determined.

```

q6_1 = ...          # G'(1)
q6_2 = ...          # G_X(t)

throws = {(i, j): Rational(1, 16) for i in
range(1, 5) for j in range(1, 5)}
M = Dist(throws, 'M', rule=max)
balls = 'RRYYY'
draws = {(i, j): Rational(1, 20) for i in
range(5) for j in range(5) if i != j}
X = Dist(draws, 'X', rule=lambda pair:
(balls[pair[0]] == 'R') + (balls[pair[1]] ==
'R'))

verify_moment("6.1 G'(1)", q6_1, Expect(M))
verify_pgf('6.2', q6_2, X, t)

```

6.3 — May 2025 TZ1 Paper 3 Q1(e)-(f), 10 marks

(The bag of 6.2.) An unbiased coin and a biased coin are tossed. The probability of obtaining a tail on the biased coin is p . Y is the total number of tails obtained from tossing both coins, and $G_Y(t) = \sum_{y=0}^2 P(Y = y)t^y$.

(e) Given that the coefficient of t^2 in $G_Y(t)$ is $\frac{1}{3}$, find **(i)** the value of p ; **(ii)** an expression for $G_Y(t)$.

Z is the sum of X and Y , and $G_Z(t) = G_X(t)G_Y(t)$.

(f) Given that $G'_Z(1) = E(Z)$, find $E(Z)$.

```
q6_3p = ...      # p
q6_3y = ...      # G_Y(t)
q6_3z = ...      # E(Z)

balls = 'RRYYY'
draws = {(i, j): Rational(1, 20) for i in
range(5) for j in range(5) if i != j}
X = Dist(draws, 'X', rule=lambda pair:
(balls[pair[0]] == 'R') + (balls[pair[1]] ==
'R'))
p = symbols('p')
Y = Dist(Dist({0: Rational(1, 2), 1:
Rational(1, 2)}, 'fair') + Dist({0: 1 - p, 1:
p}, 'biased'), 'Y')
top = Eq(P(Y == 2), Rational(1, 3))

verify_letters('6.3(e)(i)', q6_3p, p, [Y],
[top])
verify_pgf('6.3(e)(ii)', q6_3y, Y, t,
given=top, unknowns=p)
verify_moment('6.3(f)', q6_3z, Expect(X + Y),
given=top, var=p)
```



Solutions

1.1 $4.5a + 0.307 = 1$, $a = \boxed{0.154}$;
probabilities 0.231, 0.308, 0.281, 0.154, 0.026,
mode $\boxed{2}$.

1.2 $0.88 + k - 2k^2 = 1$ gives
 $\boxed{2k^2 - k + 0.12 = 0}$; $k = 0.2$ or 0.3 , and $k =$
 0.2 makes $P(X = 1) = -0.08 < 0$: $\boxed{k = 0.3}$.

1.3 $\frac{7}{2}p = 1$, $\boxed{p = \frac{2}{7}}$. $\boxed{0 \leq r \leq 1}$; $q = \frac{1-r}{3}$, so
 $\boxed{0 \leq q \leq \frac{1}{3}}$.

1.4 $0.6 - 2a \geq 0$, $3a \geq 0$, $0.4 - a \geq 0$:
 $\boxed{0 \leq a \leq 0.3}$.

2.1 $0.231 + 0.616 + 0.843 + 0.616 + 0.13 =$
 $\boxed{2.44}$ (2.436 exact).

2.2 With $k = 0.3$: $0 + 0.02 + 0.92 + 0.33 =$
 $\boxed{1.27}$.

$$\mathbf{2.3} \quad E(X) = 8p = \boxed{\frac{16}{7}}. \quad E(Y) = 6q + 4r = 4 - 6q, \text{ so } \boxed{2 \leq E(Y) \leq 4}.$$

$$\mathbf{2.4} \quad \frac{34}{40} = \boxed{0.85}; \quad \frac{60}{40} = \boxed{1.5}; \quad \frac{1000 \times 1.5}{10} = \boxed{150}.$$

$$\mathbf{2.5} \quad \frac{1+6+15+28}{16} = \boxed{3.125}.$$

$$\mathbf{2.6} \quad 0.2 + 0.64 + 0.864 + 0.6144 + 0.192 = 2.5104 \quad \checkmark$$

$$\mathbf{3.1} \quad p + q = 0.6 \quad \text{and} \quad p + 3q = 1: \quad \boxed{p = 0.4, \quad q = 0.2}.$$

$$\mathbf{3.2} \quad k^3 - k^2 - 2k + 0.7 = 0; \quad \text{only} \quad k = 0.315870\dots \text{ keeps every cell in } [0, 1], \text{ and } a = 1 - k - k^2 - k^3 = \boxed{0.553}.$$

$$\mathbf{3.3} \quad p + q = 7 \quad \text{and} \quad p + 2q + 38 = 48: \quad \boxed{p = 4, \quad q = 3}.$$

$$\mathbf{3.4} \quad q = 95 - 2p \text{ from the mean; the variance gives } p = 45 \text{ or } p = -10, \text{ and } p \in \mathbb{Z}^+: \quad \boxed{p = 45, \quad q = 5}.$$

$$\mathbf{3.5} \quad P(X < Y) = 3p(q + r) = \frac{6}{7}(q + r) = \frac{1}{2}, \text{ with } 3q + r = 1: \quad q = \frac{5}{24}, \quad r = \frac{3}{8}, \quad E(Y) = 4 - 6q = \boxed{\frac{11}{4}}.$$

4.1 $\text{Var}(2 - X) = \text{Var}(X) = E(X^2) - E(X)^2 = 4.4 - 4 = \boxed{0.4}.$

4.2 $10 \times 3 = \boxed{30}.$

4.3 $a - 4.723b = 100, \quad a - 2.25b = 150: \quad b = 20.2183\dots, \quad a = 195.491\dots, \quad \boxed{a = 195, b = 20}.$
 $\text{Var}(P) = b^2 \times 0.906 = 370.356\dots = \boxed{370}.$

4.4 $E(Y^2) = 7.4896, \quad \text{Var}(Y) = 7.4896 - 2.5104^2 = \boxed{1.19}.$

5.1 (i) $x - 1$ actions not boosted, then one boosted. (ii) $\boxed{\sum_{x=1}^{\infty} xp(1-p)^{x-1}}.$ (c)(ii)

$a = p, \quad r = 1 - p: \quad \frac{p}{p^2} = \frac{1}{p}. \quad \text{(d)}$
 $\boxed{E(X) = 10, \text{Var}(X) = 90}.$

5.2 (i) $p = \frac{1}{2.5104} = \boxed{0.398}.$ (ii) $\frac{1-p}{p^2} = \boxed{3.79}.$
 (iii) The means are equal and $\text{Var}(Y) = 1.19 < 3.79$: the second model.

6.1 $G'(1) = \frac{1+6+15+28}{16} = 3.125 = E(M) \checkmark$

$$\mathbf{6.2} \quad P(X=0) = \frac{3}{5} \cdot \frac{2}{4} = \frac{3}{10}, \quad P(X=1) = \frac{2}{5} \cdot \frac{3}{4} + \frac{3}{5} \cdot \frac{2}{4} = \frac{3}{5}, \quad P(X=2) = \frac{2}{5} \cdot \frac{1}{4} = \frac{1}{10}.$$

$$\mathbf{6.3} \quad (\text{e})(\text{i}) \quad \frac{1}{2}p = \frac{1}{3}, \quad \boxed{p = \frac{2}{3}}. \quad (\text{ii})$$

$$\boxed{G_Y(t) = \frac{1}{6} + \frac{1}{2}t + \frac{1}{3}t^2}. \quad (\text{f}) \quad E(Z) = G'_X(1) +$$

$$G'_Y(1) = \frac{4}{5} + \frac{7}{6} = \frac{59}{30} = \boxed{1.97}.$$